Started on	Wednesday, 30 September 2026, 10:07 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 10:20 PM
Time taken	13 mins 39 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as “No”.

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value “x”)

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>
void findSumPair(int arr[], int low, int high, int x){
    if(low>=high){
        printf("No\n");
        return;
    }
    int current_sum = arr[low]+arr[high];
    if(current_sum == x){
        printf("%d\n", arr[low]);
        printf("%d\n", arr[high]);
        return;
    }
    else if(current_sum < x){
        findSumPair(arr, low+1, high,x);
    }
    else{
        findSumPair(arr, low, high-1, x);
    }
}
int main(){
    int n,x;
    if(scanf("%d",&n)!=1) return 0;
    int *arr = (int *)malloc(n*sizeof(int));
    for(int i=0;i<n;i++){
        if(scanf("%d",&arr[i])!=1) return 0;
    }
    if(scanf("%d",&x)!=1) return 0;
```

```
29      findSumPair(arr,0,n-1,x);
30      free(arr);
31      return 0;
32  }
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.